

AI in Clinical Practice – Blackboard Course Shell

Diplomado en Inteligencia Artificial en Ciencias de la Salud

Format for upload: Each “Module” below = one Blackboard Content Folder (Week). Paste the Overview into the folder’s landing item, readings as Web Links, and the Discussion Prompt into a Discussion Board thread. **Adaptation note:** All “In-Hospital Practical Sandbox” hours have been converted to **Virtual Clinical Sandbox (VCS)** activities using synthetic/de-identified data and simulated patients, since students in this cohort do not have hospital or clinical-lab access. No real patient data is used anywhere in this course.

VIRTUAL CLINICAL SANDBOX (VCS) – TOOL KIT

(Post this once as a standing Blackboard page; every module references it.)

Tool	Purpose	Access
Synthea	Fully synthetic patient records (no real PHI)	Free, open-source
MIMIC-CXR / MIMIC-IV (demo/de-identified subset)	Radiology + ICU data for diagnostic-AI exercises	Free registration required
OpenEMR or GNU Health (sandbox mode)	Practice EHR environment	Free, self-hosted or instructor-hosted instance
Standardized-patient roleplay (peer-to-peer or AI patient-simulator)	History-taking + ambient scribe practice	Zoom/Teams breakout rooms
AI ambient scribe (trial/sandbox license)	Note generation practice	Vendor sandbox account (no PHI permitted)
Virtual OSCE platform (e.g., Oxford Medical Simulation, Body Interact, or instructor-built station)	Mock OSCE & capstone shift	Institutional license or free trial

Standing safety rule for all VCS activities: Only synthetic or fully de-identified public datasets may be used. Students must never upload real patient information to any AI tool used in this course.

MONTH 1 – ADMIN TIME SAVINGS & THE AI FOUNDATION

Module – Week 1 (Oct 1-7): Introduction to AI in

Medicine

Learning Objectives

- Define Human-in-the-Loop (HITL) and explain why it anchors the whole course
- Distinguish supervised vs. generative AI models at a conceptual level
- Define “hallucination” in a clinical-AI context

Content Overview (Virtual, 3 hrs)

- Course orientation; landscape of clinical AI tools
- HITL principle walkthrough
- ML literacy basics

Virtual Clinical Sandbox Activity (3 hrs)

- Set up VCS accounts (Synthea, sandbox EHR, ambient-scribe trial)
- Baseline exercise: time a mock documentation task *without* AI, for later comparison

Required Readings

- NIH/AAMC Clinical AI Competency Model
- HITL workflow overview

Discussion Board Prompt (Q1.1) > A physician signs an AI-scribed note without reading it, missing a mistranscribed chest-pain denial. What systemic failure does this illustrate — and how would VCS practice this week have caught it?

Module — Week 2 (Oct 8-14): Prompt Engineering — The “Four Rs”

Learning Objectives

- Apply the AAFP Four Rs framework (Request, Reason, References, Revise) to clinical prompts
- Identify common prompt-writing failure modes

Content Overview (Virtual, 3 hrs)

- Four Rs framework lecture + live-prompting workshop

Virtual Clinical Sandbox Activity (3 hrs)

- Pair up for standardized-patient roleplay (one student as patient using a Synthea case card, one as clinician)
- Run the ambient scribe on the recorded roleplay; each student submits one “before/after” prompt revision

Required Readings

- AAFP “Four Rs” framework
- Prompt literacy in clinical training

Discussion Board Prompt (Q1.2) > Which prompt structure best reflects the Four Rs, and why do the other options fail to build in a verification step?

Module — Week 3 (Oct 15-21): Ambient Scribing in Practice

Learning Objectives

- Categorize ambient-scribe error types (mistranscription, omission, hallucinated detail)
- Explain the German GP digital-literacy gap as a training driver

Content Overview (Virtual, 3 hrs)

- Error taxonomy deep dive; German digitalization context

Virtual Clinical Sandbox Activity (3 hrs)

- Full mock-clinic day: 3-4 synthetic-patient roleplay encounters of varying complexity, using the ambient scribe throughout
- Maintain a “catch log” of every AI correction made

Required Readings

- German GP digital literacy gap (compliance guide)
- Clinical AI evidence base

Discussion Board Prompt (Q1.3) > Why does the curriculum mandate structured onboarding for AI literacy rather than self-study, specifically in the German primary-care context?

Module — Week 4 (Oct 22-28): Month 1 Synthesis & Checkoff

Learning Objectives

- Distinguish “time saved” from “output trustworthiness”

Content Overview (Virtual, 3 hrs)

- Class review of catch logs; efficiency-vs-trust discussion

Virtual Clinical Sandbox Activity (3 hrs) — CHECKOFF

- Independently complete one full ambient-scribed synthetic encounter with a

signed, corrected note; submit via Blackboard assignment for instructor grading

Discussion Board Prompt (Q1.4) > A resident says the scribe “saved two hours so I trust it completely now.” What’s wrong with that reasoning?

Milestone: Month 1 Competency Checkoff — pass/fail gate to Month 2

📅 MONTH 2 — AI-DRIVEN DIAGNOSTICS & THE SAFETY OVERRIDE

Module — Week 5 (Oct 29-Nov 4): Critical Appraisal of AI Diagnostics

Learning Objectives

- Explain the dissociation between board-exam AI performance and bedside reasoning
- Identify “confident hallucinations” in diagnostic AI output

Content Overview (Virtual, 3 hrs)

- Evidence review of AI benchmark vs. real-world performance gaps

Virtual Clinical Sandbox Activity (3 hrs)

- Observe (without acting) a diagnostic-AI demo run against MIMIC-CXR de-identified sample images; log confidence scores vs. actual findings

Required Readings

- AI board-exam vs. bedside reasoning dissociation
- Clinical AI evaluation methods

Discussion Board Prompt (Q2.4) > An LLM aces board questions but fails multi-step bedside reasoning. What does this dissociation teach us about evaluating clinical AI?

Module — Week 6 (Nov 5-11): German Quality Pillars

Learning Objectives

- Differentiate Functionality, Robustness, and Comprehensibility (Explainable AI) failure modes

Content Overview (Virtual, 3 hrs)

- Three-pillar framework; GDPR “right to explanation”

Virtual Clinical Sandbox Activity (3 hrs)

- Case-review exercise: sort a set of de-identified CDSS malfunction write-ups by which pillar was violated

Required Readings

- German quality pillars
- GDPR/XAI compliance

Discussion Board Prompt (Q2.2) > A CDSS gives a recommendation with no visible reasoning trace. Which pillar does this violate, and why isn't it a Functionality problem instead?

Module — Week 7 (Nov 12-18): The Sandbox Override Exercise

Learning Objectives

- Execute a manual override on a flawed AI diagnostic output
- Recognize bias effects from underrepresented training data

Content Overview (Virtual, 3 hrs)

- Override methodology; bias case discussion

Virtual Clinical Sandbox Activity (3 hrs) — SEEDED EXERCISE

- Instructor seeds anomalous/biased synthetic cases into the VCS diagnostic tool; students must detect the error, articulate why it's implausible, and document a manual override

Discussion Board Prompt (Q2.1, Q2.3) > Walk through your override decision from this week's seeded case. What clinical detail tipped you off that the AI was wrong?

Module — Week 8 (Nov 19-25): Month 2 Synthesis & Checkoff

Content Overview (Virtual, 3 hrs)

- Class debrief of override outcomes and near-misses

Virtual Clinical Sandbox Activity (3 hrs) — CHECKOFF

- Independently detect and override one new seeded CDSS error under observation; submit documentation

Milestone: Month 2 Competency Checkoff — pass/fail gate to Month 3

🏠 MONTH 3 — CLINICAL CARE PATHWAYS & DATA SOVEREIGNTY

Module — Week 9 (Nov 26-Dec 2): AI-Enhanced Care Planning

Learning Objectives

- Combine NLP-extracted problem lists with a standardized nursing care model
- Apply HITL review specifically to nursing workflows

Content Overview (Virtual, 3 hrs)

- Care-planning AI walkthrough

Virtual Clinical Sandbox Activity (3 hrs)

- Draft an AI-assisted care plan from a Synthea synthetic patient record; revise against a simulated bedside presentation (peer roleplay)

Required Readings

- AI-enhanced clinical encounters

Discussion Board Prompt (Q3.2) > What must a nurse verify before finalizing an AI-drafted care plan, and why can't NLP extraction alone be trusted as "deterministic"?

Module — Week 10 (Dec 3-9): GDPR & Data Sovereignty

Learning Objectives

- Explain why cross-border cloud AI use is restricted for identifiable EU health data
- Articulate the rationale for local/open-weight AI deployment

Content Overview (Virtual, 3 hrs)

- GDPR case study; sovereignty rationale

Virtual Clinical Sandbox Activity (3 hrs)

- Map a data-flow diagram for the VCS tools used this term; mark every point where data would leave institutional control if this were a real hospital system

Required Readings

- EU GDPR healthcare compliance guide

- Local/open-weight deployment rationale

Discussion Board Prompt (Q3.1, Q3.4) > Why would a German hospital block a US cloud LLM API for care-plan generation, and why might it prefer an on-premises open-weight model instead?

Module — Week 11 (Dec 10-16): Anonymization & Secondary Data Use

Learning Objectives

- Differentiate anonymization vs. pseudonymization
- Describe the DiGA pathway as a compliance model

Content Overview (Virtual, 3 hrs)

- Data-minimization principles; DiGA overview

Virtual Clinical Sandbox Activity (3 hrs)

- Extract an anonymized dataset from Synthea records for a mock research request; document the anonymization steps taken

Required Readings

- DiGA (Digital Health Applications) pathway

Discussion Board Prompt (Q3.3) > What must be true of a dataset before it's used for external research, and how does this week's Synthea exercise satisfy that standard?

Module — Week 12 (Dec 17-23): Month 3 Synthesis & Checkoff

Content Overview (Virtual, 3 hrs)

- Local EHR-embedded AI modules review

Virtual Clinical Sandbox Activity (3 hrs) — CHECKOFF

- Produce one GDPR-compliant, anonymized care-plan data extract from a synthetic record; submit for grading

Milestone: Month 3 Competency Checkoff — pass/fail gate to Month 4 (*Suggest a scheduled break after this module before Week 13 resumes.*)

▣ MONTH 4 — ETHICOLEGAL AUTONOMY & THE FINAL CHECKOFF

Module — Week 13 (Dec 24-30): Liability Mapping

Learning Objectives

- Identify who bears responsibility when an AI recommendation is wrong

Content Overview (Virtual, 3 hrs)

- Liability case conference (de-identified adverse-event report)

Virtual Clinical Sandbox Activity (3 hrs)

- Small-group analysis of the case; each group presents a liability map (clinician/vendor/institution)

Required Readings

- NIH/AAMC liability & practice-based learning

Discussion Board Prompt (Q4.3) > If a clinician follows a wrong AI recommendation, who bears primary responsibility under EU/German frameworks, and why doesn't following the AI's advice shift that responsibility?

Module — Week 14 (Dec 31-Jan 6): EU AI Act Risk Tiers

Learning Objectives

- Classify a clinical AI tool by EU AI Act risk tier

Content Overview (Virtual, 3 hrs)

- Risk-tier structure lecture

Virtual Clinical Sandbox Activity (3 hrs)

- Tier-classification exercise using all VCS tools used this term (scribe, CDSS demo, care-planning tool)

Required Readings

- EU AI Act overview

Discussion Board Prompt (Q4.4) > Classify the ambient scribe, the diagnostic CDSS, and the care-planning NLP tool by EU AI Act risk tier. Justify each.

Module — Week 15 (Jan 7-13): Informed Consent & Transparency

Learning Objectives

- Disclose AI involvement to a patient appropriately

Content Overview (Virtual, 3 hrs)

- Consent-conversation scripting practice

Virtual Clinical Sandbox Activity (3 hrs)

- Standardized-patient video roleplay requiring AI-use disclosure; peer-graded against a rubric

Required Readings

- EU AI Act & patient consent

Discussion Board Prompt (Q4.1) > A patient asks if AI was used in their diagnosis. Draft the disclosure response you'd actually say out loud, then explain why it satisfies transparency obligations.

Module — Week 16 (Jan 14-20): Capstone Preparation

Content Overview (Virtual, 3 hrs)

- Solo Pilot simulation walkthrough + grading rubric review
- Integration review across Months 1-4

Virtual Clinical Sandbox Activity (3 hrs)

- Ungraded dry-run capstone shift with instructor feedback

Discussion Board Prompt (Q4.2) > During your dry-run, did the AI imaging tool ever disagree with your own read? Walk through how you resolved it.

Module — Week 17 (Jan 21-27): Month 4 Synthesis & Checkoff

Content Overview (Virtual, 3 hrs)

- Final review; open Q&A

Virtual Clinical Sandbox Activity (3 hrs) — CHECKOFF

- Graded mock disclosure conversation + risk-tier classification task

Milestone: Month 4 Competency Checkoff — pass/fail gate to Consolidation

■ MONTH 5 — CONSOLIDATION & MOCK OSCE

Module — Week 18 (Jan 28-Feb 3): Integration

Station Design

Content Overview (Virtual, 3 hrs)

- Build integrative cases combining all four competency areas

Virtual Clinical Sandbox Activity (3 hrs)

- Students design and pilot one integration station on a peer group, using synthetic data throughout
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Module — Week 19 (Feb 4-10): Mock OSCE Round 1

Virtual Clinical Sandbox Activity (3 hrs)

- Timed virtual OSCE stations via video platform, instructor-scored

Discussion Board Prompt (Q5.1) > Walk through every HITL checkpoint from Months 1-4 you applied during your OSCE station.

Module — Week 20 (Feb 11-17): Mock OSCE Round 2 + Remediation

Virtual Clinical Sandbox Activity (3 hrs)

- Repeat stations for students needing remediation; individual feedback sessions

Discussion Board Prompt (Q5.2) > Using one example from your OSCE stations, distinguish an “automation bias” failure from an “explainability failure.”

Module — Week 21 (Feb 18-24): Consolidation Wrap-Up

Content Overview (Virtual, 3 hrs)

- Full-class synthesis; open forum

Virtual Clinical Sandbox Activity (3 hrs)

- Free-practice sandbox time, student-directed

Milestone: Consolidation Sign-Off — readiness gate to Capstone

■ MONTH 6 — SOLO PILOT CAPSTONE & CERTIFICATION

Module — Week 22 (Feb 25-Mar 3): Capstone Shift 1

Virtual Clinical Sandbox Activity (6 hrs, full shift)

- Full simulated shift using synthetic EHR + roleplay patient + synthetic imaging: intake (AI scribing) → diagnosis (AI imaging validation) → treatment planning, graded against the “Confidence Factor” checklist

Module — Week 23 (Mar 4-10): Capstone Shift 2 + Individual Debrief

Virtual Clinical Sandbox Activity (6 hrs)

- Second graded shift, different synthetic case mix; one-on-one debrief with faculty

Module — Week 24 (Mar 11-17): Remediation / Retake Window

Virtual Clinical Sandbox Activity

- Open sandbox access for students needing a final practice pass

Module — Week 25 (Mar 18-24): Certification Ceremony & Course Close

Content Overview (Virtual, 3 hrs)

- Final exam review, course evaluation

Virtual Clinical Sandbox Activity (3 hrs)

- Final graded checkoff; AI Competency Certificate awarded

Milestone: AI Competency Certificate issued (NIH/AAMC + German/EU dual-compliant)

Blackboard Upload Checklist

- ☐ Create one Content Folder per week (25 total), titled “Week N: [Theme]”
- ☐ Paste Learning Objectives + Content Overview into each folder’s landing item
- ☐ Add Required Readings as Web Links (verify each URL is live before publishing)
- ☐ Create a Discussion Board forum per week using the listed prompt
- ☐ Create graded Assignments for each Checkoff week (4, 8, 12, 17, 21, 25) with

a rubric attached

- ☐ Post the VCS Tool Kit page once, and link to it from every week's sandbox activity
- ☐ Set module release dates to match the calendar above (or adjust uniformly if your term starts on a different date)